Interpretable Prediction of Breast Cancer Therapy Response

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Group 10

Outline

Project Overview

Scientific Question:

Is it possible to predict the response of breast cancer patients to neoadjuvant chemotherapy by integrating multi-omics data and using machine learning methods with Shapley-Based explanations?

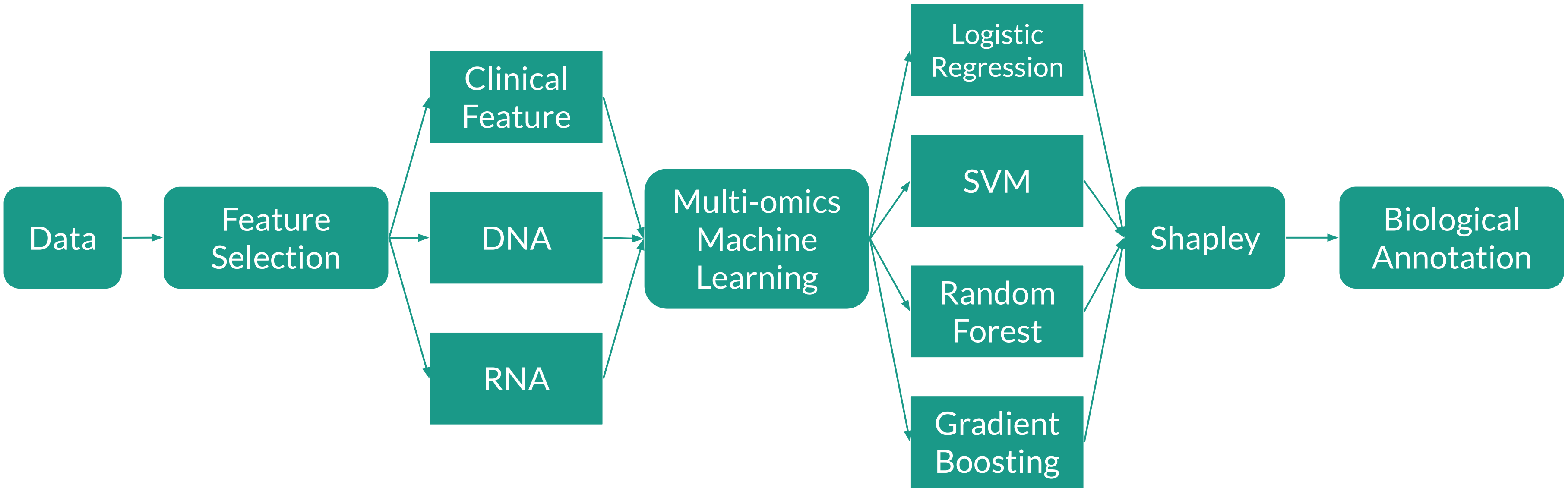
Motivation:

- pCR is a strong surrogate marker for long-term survival
- Most patients do not achieve pCR, and clinical factors alone are not sufficient for prediction

Multi-omics layer: Clinical, DNA, RNA

Biological Background

- **Neoadjuvant chemotherapy** is commonly given before surgery to shrink tumors and assess treatment sensitivity
- **Pathological complete response (pCR)**: the absence of invasive cancer in breast and lymph nodes after therapy
- **Residual Cancer Burden (RCB)** score: measures the extent of residual disease and is assessed at surgery.
- **Inspirational Paper**: Sammut et al., Nature 2022, “Multi-omic machine learning predictor of breast cancer therapy response”



Project Overview Figure

Key Innovations

Key Innovations:

- Customized **feature selection** strategies across omics layers
- Broader exploration of **machine learning models**
- Application of **SHAP** to explain model predictions at the feature and patient level.
- In-depth **biological annotation** of SHAP-selected genes to link prediction to mechanistic understanding.

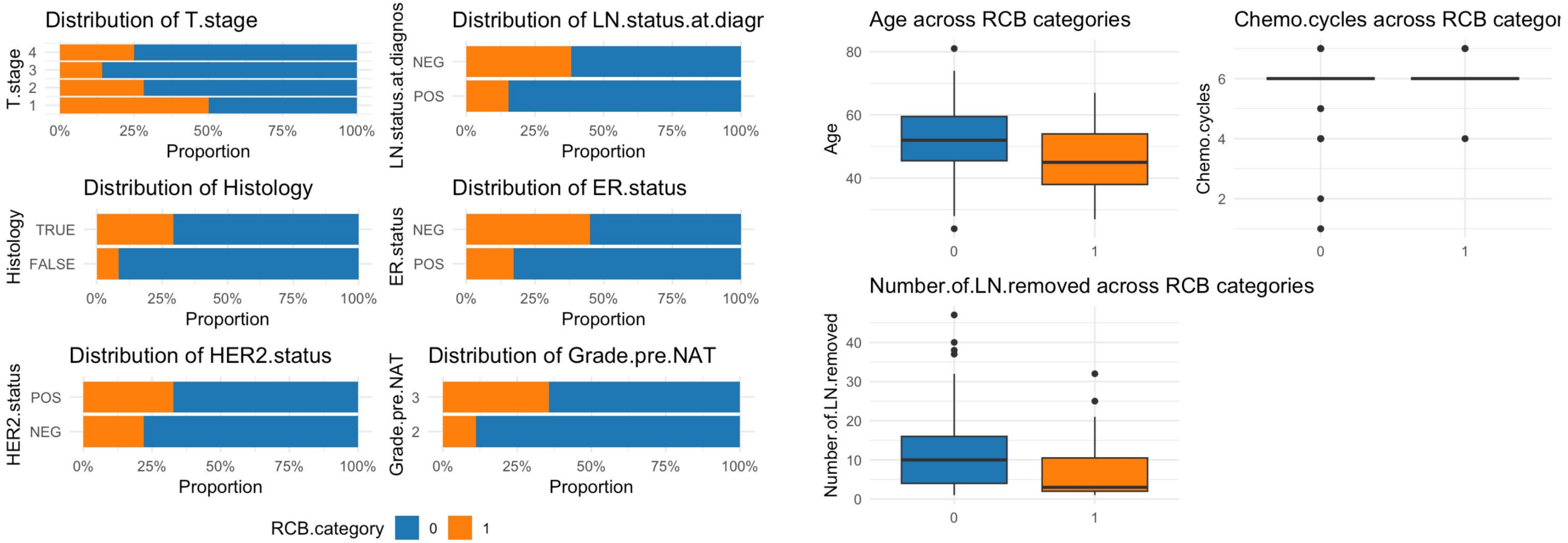
01 Data

- ➢ Multi-omics data including clinical, DNA, RNA features.
- ➢ (Preprocessed from Sammut et al., Nature 2022 GitHub repository)

Clinical Data Overview

- 168 total patient samples
- **161** of **168** samples have **RCB score** (pCR,RCB I/II/III)
- Multi-omics layers: clinical, DNA, RNA
- **Preprocessing steps:**
  - Categorical variables properly factorized
  - Removed duplicates and highly correlated features
  - Excluded variables unrelated to pre-surgical conditions

```
##  [1] "Donor.ID"                    "Age"
##  [3] "T.stage"                     "LN.status.at.diagnosis"
##  [5] "Histology"                   "ER.Allred"
##  [7] "ER.status"                   "HER2.status"
##  [9] "Grade.pre.NAT"               "NAT.regimen"
## [11] "Chemo.cycles"                "aHER2.cycles"
## [13] "Surgery.type"                "LVI"
## [15] "Tumour.dimension.surgery.1"  "Tumour.dimension.surgery.2"
## [17] "Percent.cellularity"         "Percent.CIS"
## [19] "Number.of.LN.removed"        "Number.of.positive.LN"
## [21] "Max.LN.met.size"             "RCB.score"
## [23] "RCB.category"                "pCR.RD"
## [25] "PAM50"                       "iC10"
## [27] "DNA.sequenced"               "RNA.sequenced"
## [29] "Digital.pathology"
```



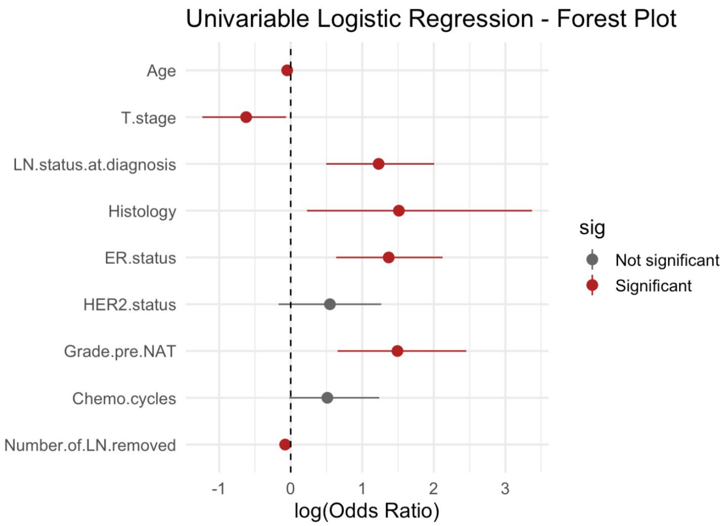
Clinical Feature Distribution by pCR/RCB Category

Univariable Logistic Regression

➢ Test the association of each single variable with pCR

Results:

- Older age → pCR odds ↓
- Larger tumor stage → pCR odds ↓
- LN-negative tumors → pCR odds ↑
- ER-negative tumors → pCR odds ↑

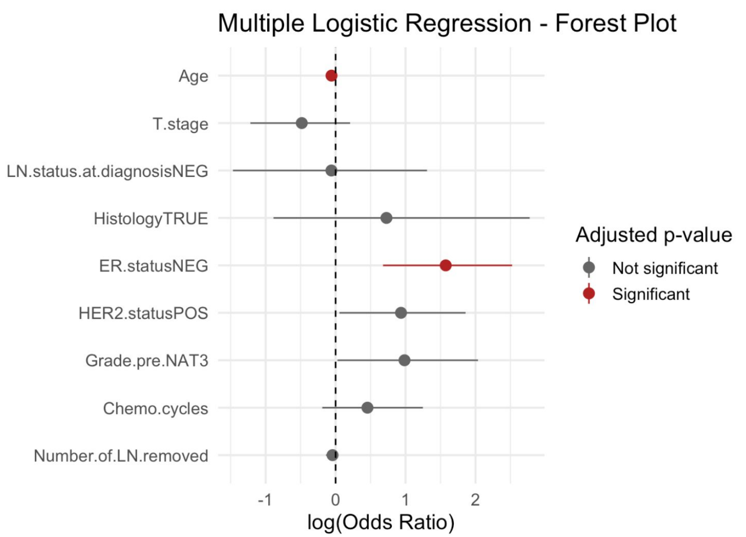


Multivariable Logistic Regression

➢ Adjust p-value (Benjamini-Hochberg)
➢ Fewer statistically significant variables

Results:

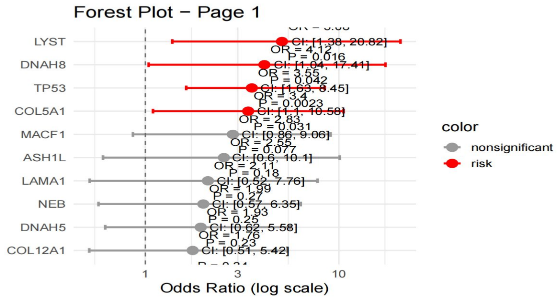
- Age and ER status remained significant after adjustment
- Other features showed weak effects (possibly due to limited sample size)
- To avoid bias from early filtering, all 9 clinical features were included in the integrative ML pipeline

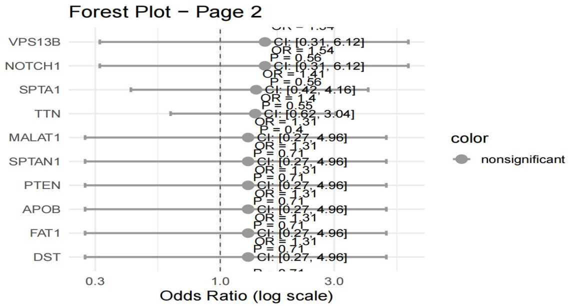


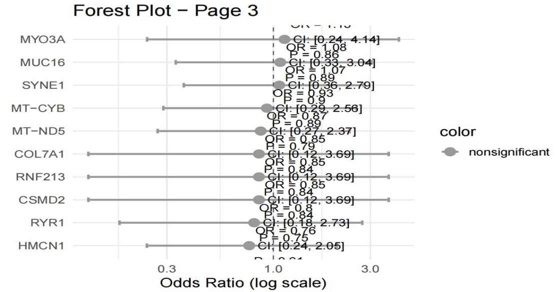
DNA Data Overview

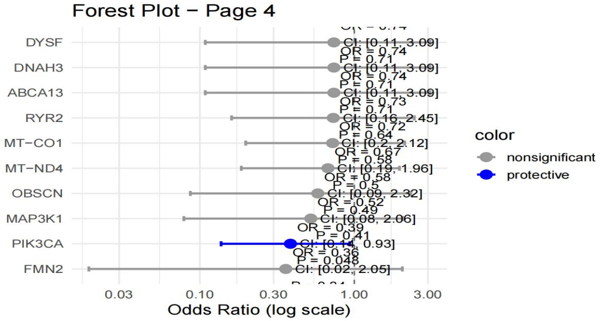
- Data: 168 samples & 16135 mutations
- first we calculate the top 5% of driver genes with the most frequent mutations
- we investigated whether mutations in these 50 genes are associated with pCR-pathological complete response.

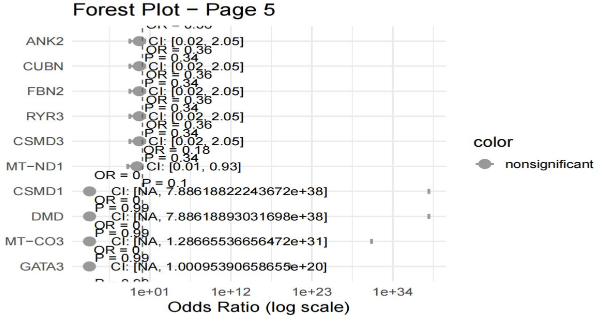
Mutations and pCR



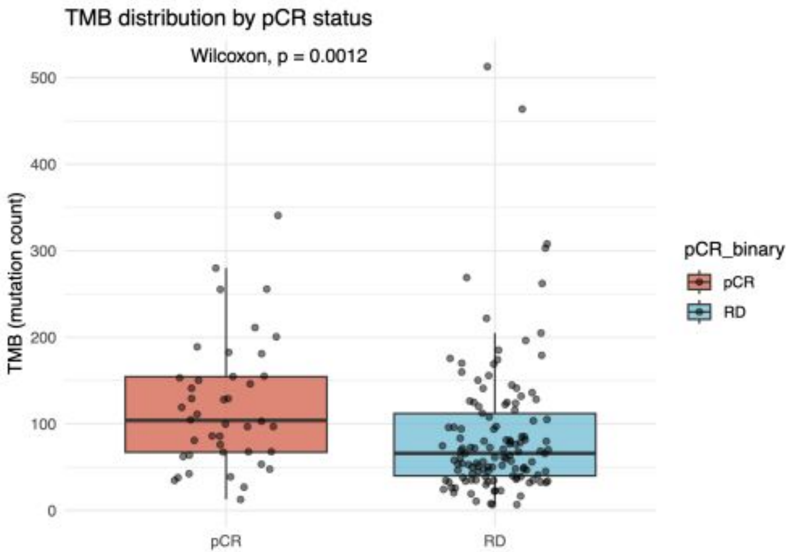


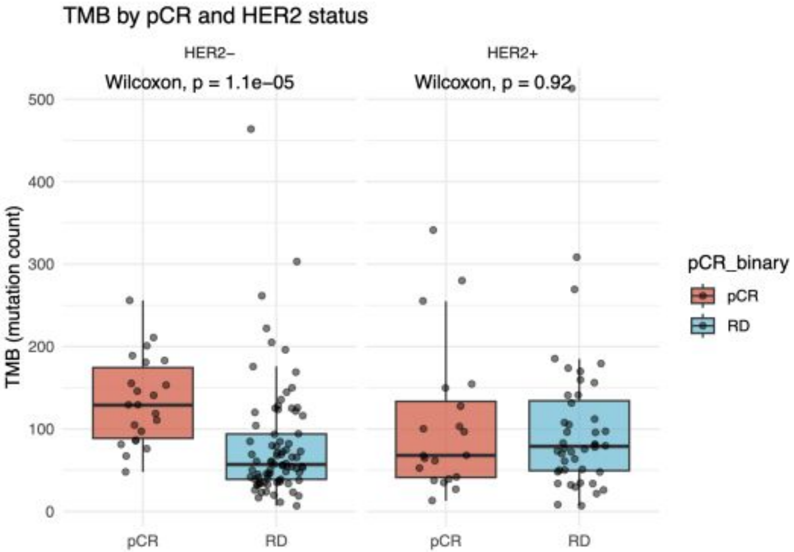




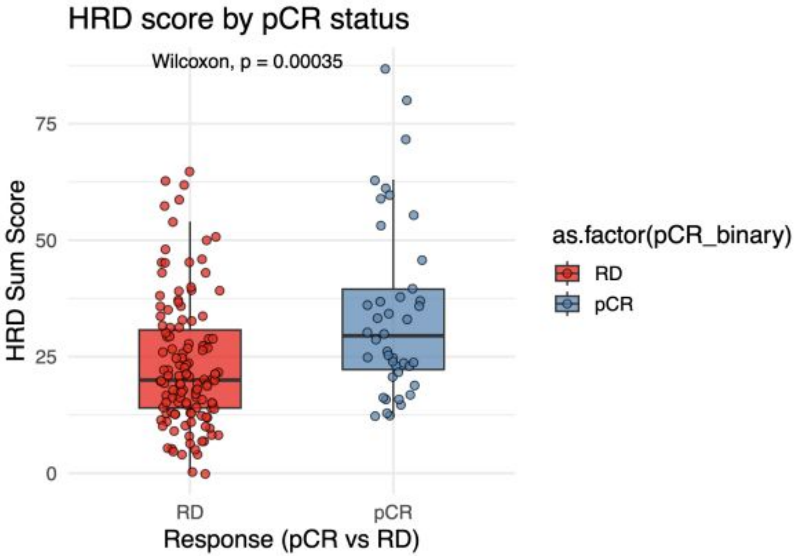


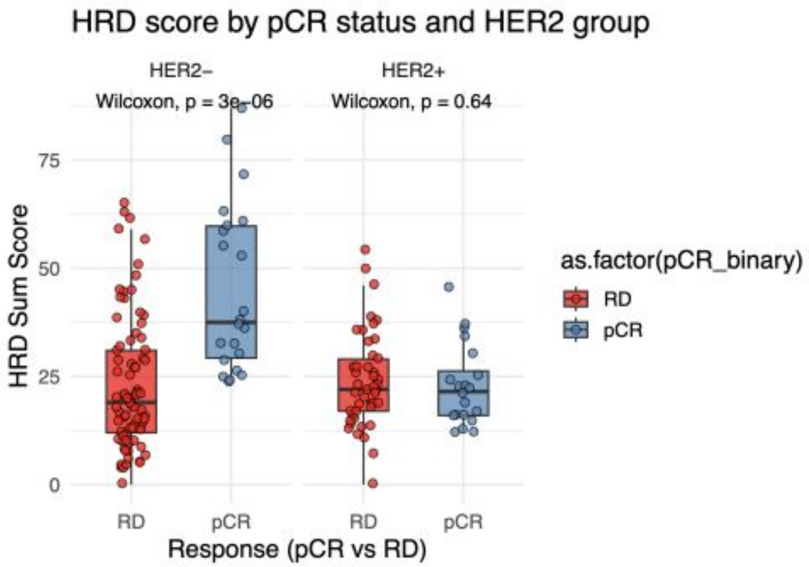
TMB (tumor mutation burden)and pCR



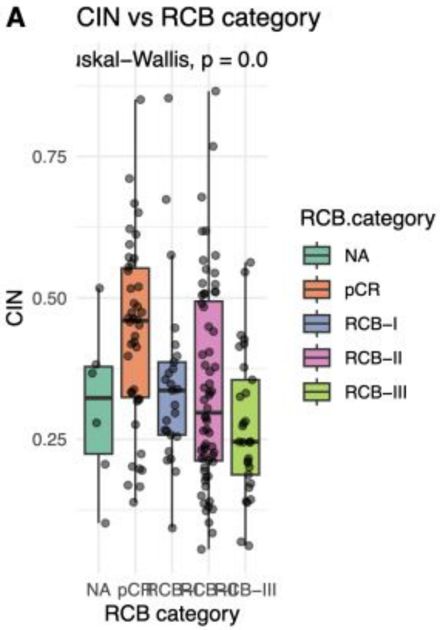


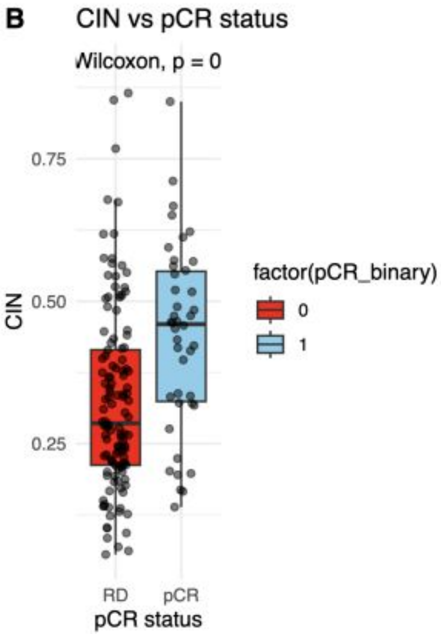
HRD(Homologous Recombination Deficiency)scores and pCR





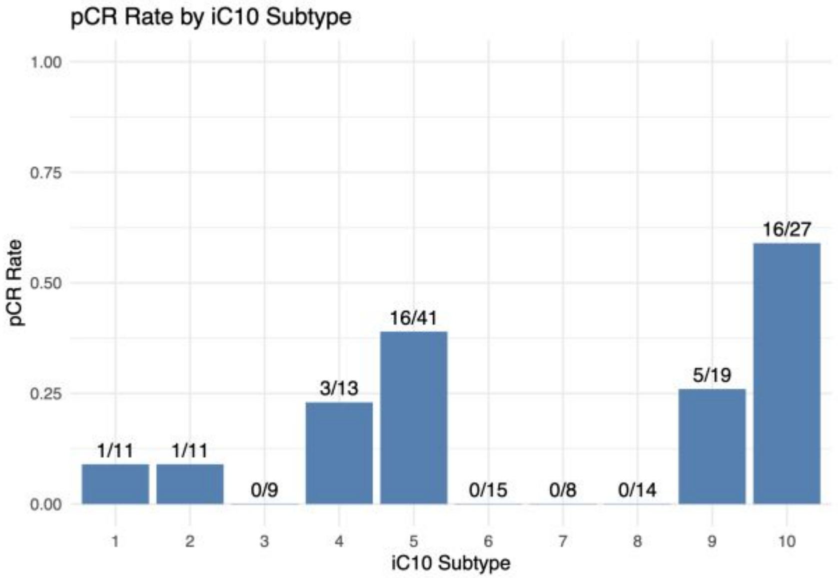
CIN(chromosomal instability)and pCR





- ➢ This is also visually evident from the boxplot: the median CIN in the pCR group is higher and the overall distribution is shifted upward.

iC10 Molecular Subtypes and pCR



- ➢ The results demonstrated marked differences:
- ➢ 1.The highest pCR rate was observed in iC10 subtype 10, reaching 59.3%.
- ➢ 2.Subtypes 5 and 9 followed, with 39.0% and 26.3%, respectively.
- ➢ 3.In contrast, subtypes 3, 6, 7, and 8 had no pCR cases, suggesting poor response.
- ➢ 4.Subtypes 1 and 2 also had low pCR rates of 9.1%.

RNA Data Overview

- 180 samples with raw RNA-seq counts
- 57 905 genes quantified per sample
- Some samples have low overall expression levels
- Dataset filtering was required before downstream analysis

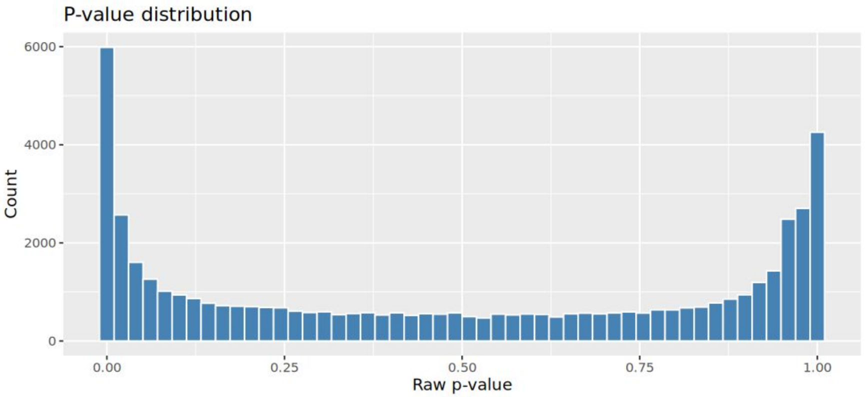
```r
# Inspect
first6_genes <- head(df)
print(first6_genes)
```

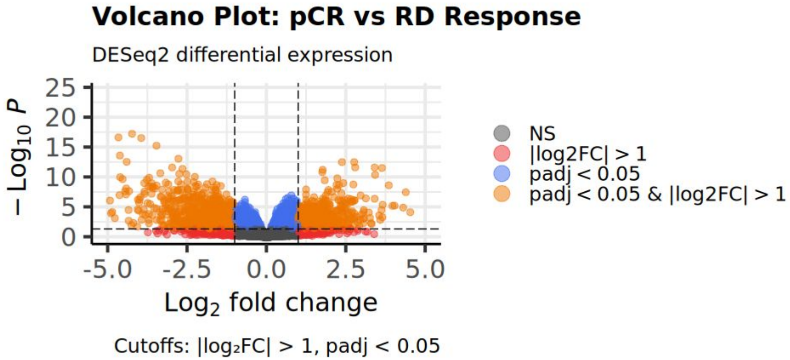
```
##                 X T001 T002 T003 T004 T005 T006 T007 T008 T009 T010 T011 T012
## 1 ENSG00000000003  625 4241  586 1038 2040  381 1089 1640 2240 1137 1450 4995
## 2 ENSG00000000005    1   30    1    5   15    2   31   30   26    4    7   90
## 3 ENSG00000000419 1176 1938 2012 1726 2355  991 2481 1106  820 1166 1662 2123
## 4 ENSG00000000457 1794 1289 2384  577 1227 1014 1361  891 1332 1044 1933 3193
## 5 ENSG00000000460  752  683 1295  404  944  321  429  413  617  454 1166 4100
## 6 ENSG00000000938  362  638 1559  842  273  372  750 1414  762  574  544  541
```

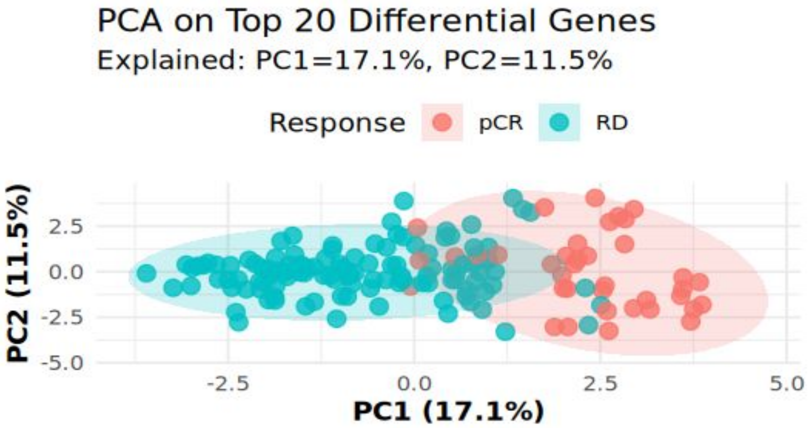
Differential expression analysis (DESeq2)

- 156 samples with raw counts
- 4 006 upregulated genes
- 3 867 downregulated genes

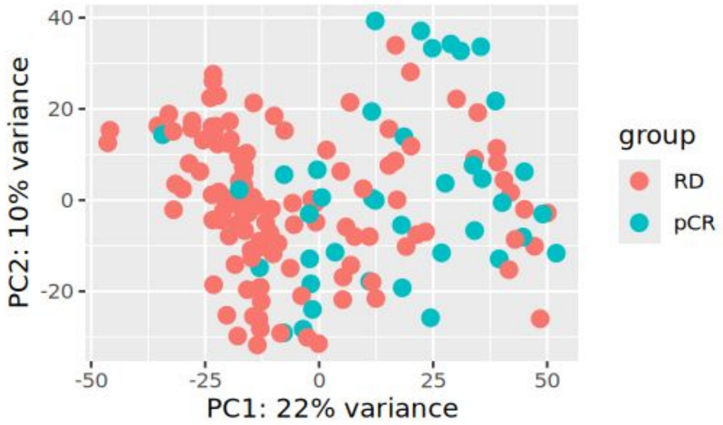
```
out of 49448 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)          : 4006, 8.1%
LFC < 0 (down)        : 3867, 7.8%
outliers [1]          : 0, 0%
low counts [2]        : 17182, 35%
(mean count < 1)
```







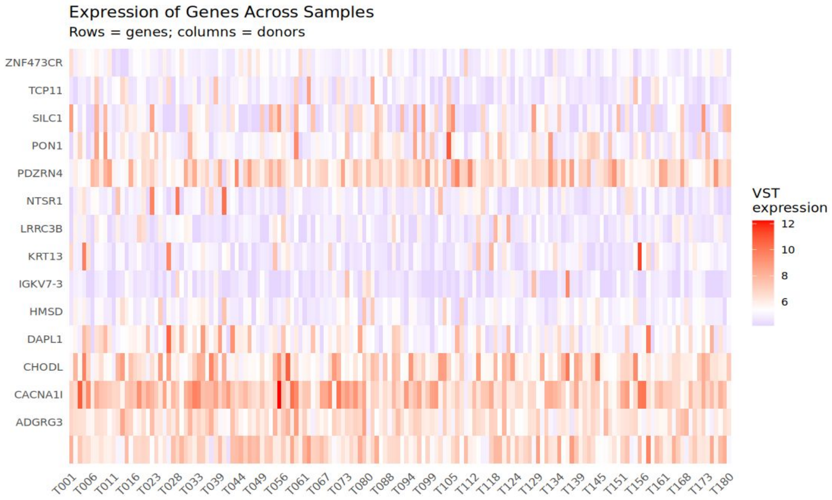
PCA on Top 20 Differential Genes
Explained: PC1=17.1%, PC2=11.5%
Response
pCR
RD
PC2 (11.5%)
PC1 (17.1%)
2.5
0.0
-2.5
-5.0
-2.5
0.0
2.5
5.0

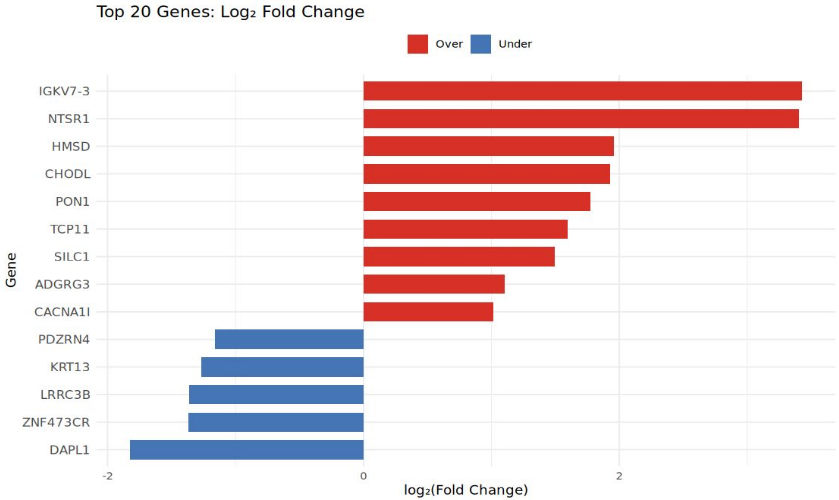


PC2: 10% variance
PC1: 22% variance
40
20
0
-20
-50
-25
0
25
50
group
RD
pCR

Feature selection using LASSO

- From 1200 genes, only 39 were highlighted as significant by LASSO

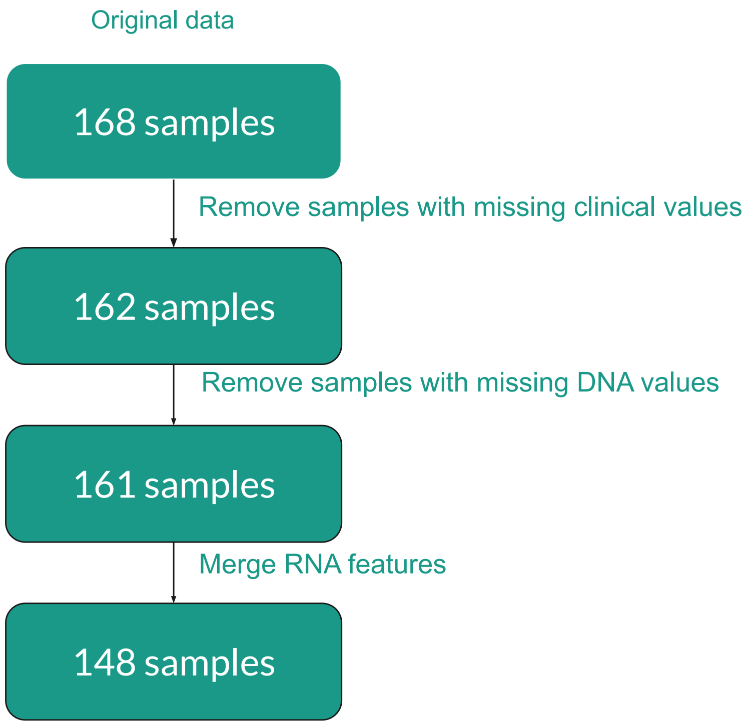




02 Integration

Data Construction & Cleaning

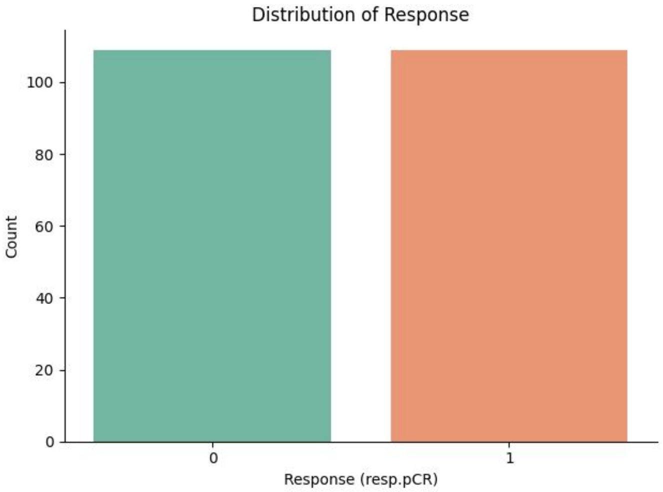
- ➢ Remove samples with missing clinical values
  - T27, T88, T106, T108, T157, T158, T165 have missing RCB values.
- ➢ Remove samples with missing DNA values
  - T18 has missing CIN value.
- ➢ Merge RNA features
  - 148 samples
  - 12 clinical features, 9 DNA features, 20 RNA features

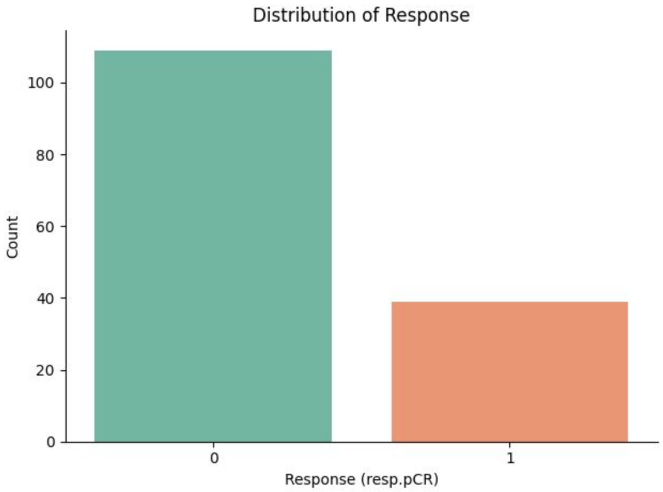


Problems

1. Pretty small dataset, no real dataset for testing -> Data Leakage
2. Imbalanced classes: 109 negatives and only 39 positives

Solution: Oversampling





SMOTE & Nested Cross-validation

Problem: Class Imbalance

This leads to high false-negative rates on the underrepresented class.

Solution: SMOTE Oversampling

General idea: Synthesizes new minority-class samples by interpolating between existing ones.

Nested CV:

Ideal for small datasets—maximizes both training and testing on every sample

Ensures each sample is used once for testing and multiple times for training, giving a less biased estimate of generalization performance.

03 Data Modeling

Models & Tuning

- **Models** used:
  - Logistic Regression
  - SVM (Linear, RBF)
  - Random Forest
  - Gradient Boosting
- General pipeline:
  - Data pre-processing
  - Feature selection
  - SMOTE for oversampling
  - Nested CV (for approach #1)

Model	Key Hyper-parameters	Range / Values
Logistic Regression	C	0.01, 0.1, 1, 10, 100
SVM	C, kernel	C: 0.01, 0.1, 1, 10, kernel: RBF, linear
Random Forest	n_estimators, max_depth	n_estimators: [100, 200], max_depth: [3, 5, None]
G.B	learning_rate, n_estimators	n_estimators": [100, 200], "clf__learning_rate": [0.01, 0.1, 0.2]

Data preprocessing & Feature Selection Strategy

Preprocessing:

- Remove any patients with any missing features or target
- Exclude any non-feature columns
- Split data into X and y datasets

Additional feature selection pipeline:

- VarianceThreshold: Remove low-variance features
- Correlation Filter: Remove highly correlated features
- SelectKBest (ANOVA F-test): Keep most significant features

Approaches

Approach #1: Use whole dataset for training

Use nested Cross-Validation with 2 loops:

- Outer loop: Evaluate model performance
- Inner loop: Tune hyperparameters via GridSearchCV

Overall: 25 variations to fit per model

+ : More data for training

- : No testing, trust on evaluated performance after CV

Approach #2: Train/test split before actual training

Use GridSearchCV only for hyperparameters tuning.

+ : Testing dataset for a more fair evaluation

- : Much less data for training given the small initial dataset

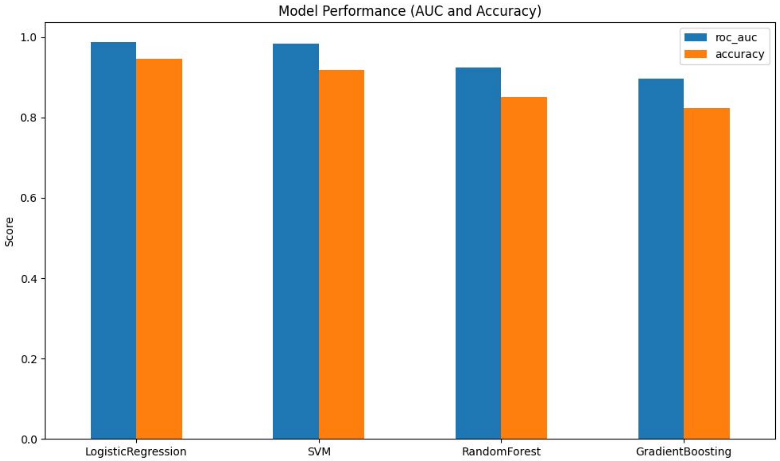
There is some data leakage, so performance can and will be **over optimistic**!

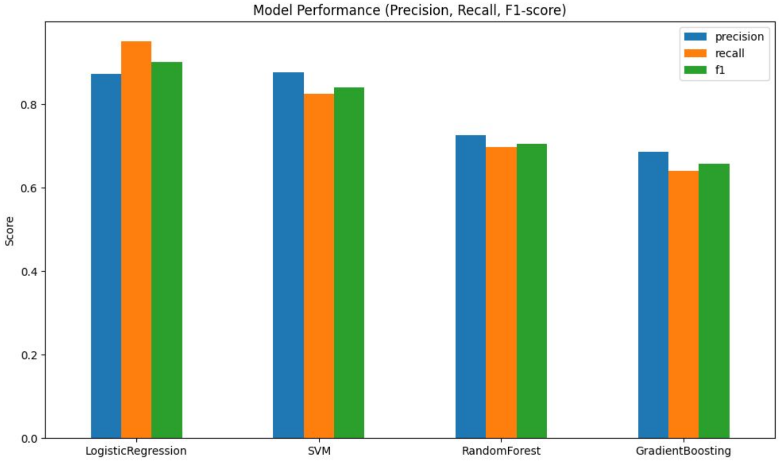
Model Evaluation Results

Reported mean scores from nested cross-validation for each model

Evaluation metrics include:

- AUC (Area Under ROC Curve)
- Accuracy
- Precision
- Recall
- F1-score





Comparison

--- Nested Cross-Validation Results ---

	roc_auc	accuracy	precision	recall	f1
LogisticRegression	0.987013	0.945977	0.872778	0.950000	0.901886
SVM	0.982792	0.919080	0.876190	0.825000	0.839426
RandomForest	0.924540	0.851264	0.725000	0.696429	0.704748
GradientBoosting	0.896104	0.824368	0.686667	0.639286	0.657143

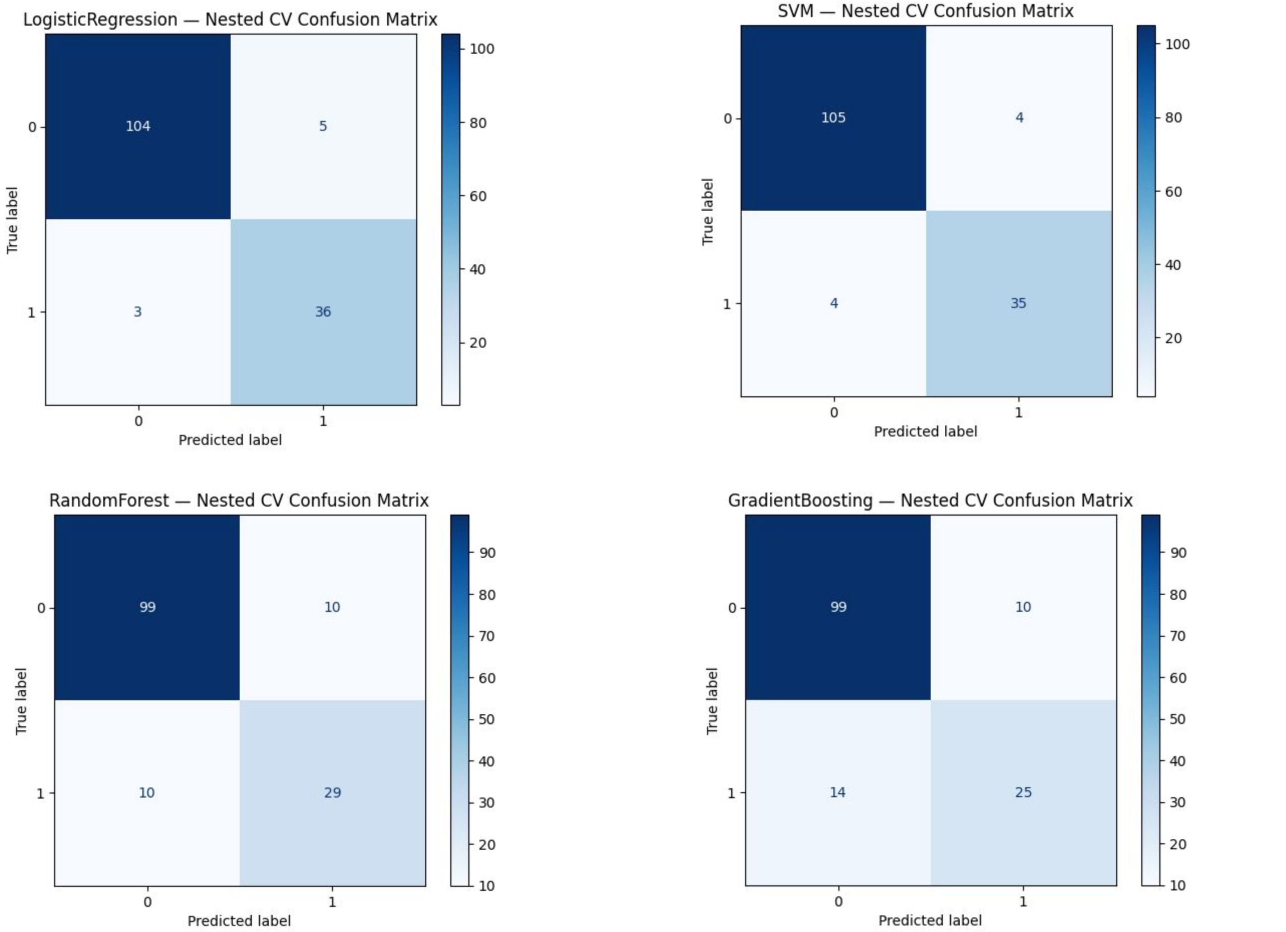
--- Train/Test Split Results ---

	roc_auc	accuracy	precision	recall	f1
LogisticRegression	0.989899	0.933333	0.846154	0.916667	0.880000
SVM	0.967172	0.911111	0.900000	0.750000	0.818182
RandomForest	0.904040	0.866667	0.750000	0.750000	0.750000
GradientBoosting	0.914141	0.866667	0.800000	0.666667	0.727273

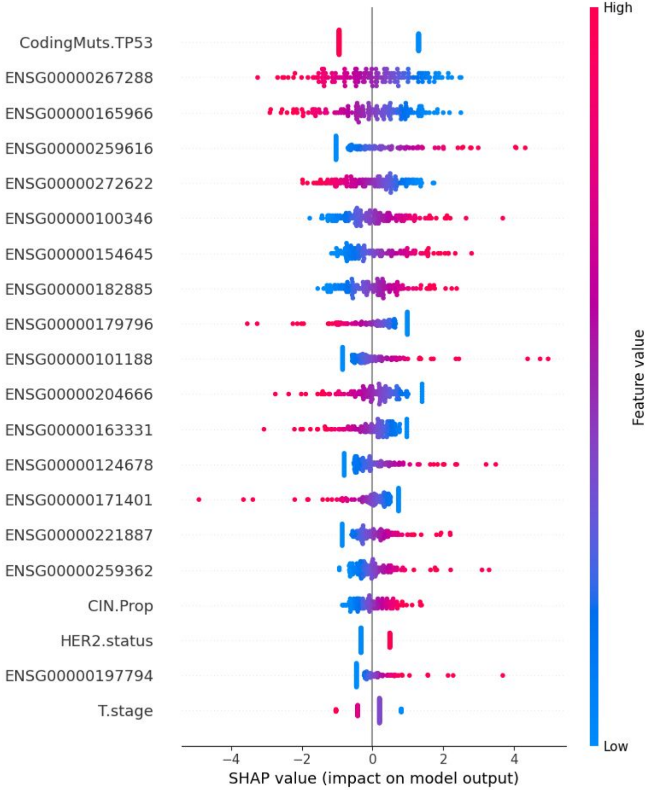
Logistic Regression and **SVM** appear to be the most promising models for this task.

Nested cross-validation provides a more reliable assessment of their expected performance on unseen data.

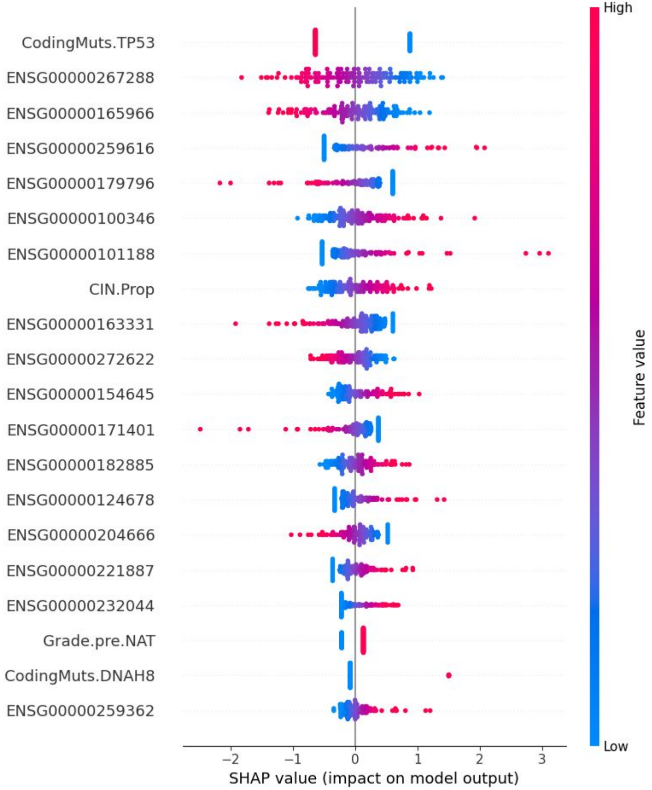
Nested CV Confusion Matrix



SHAP Analysis



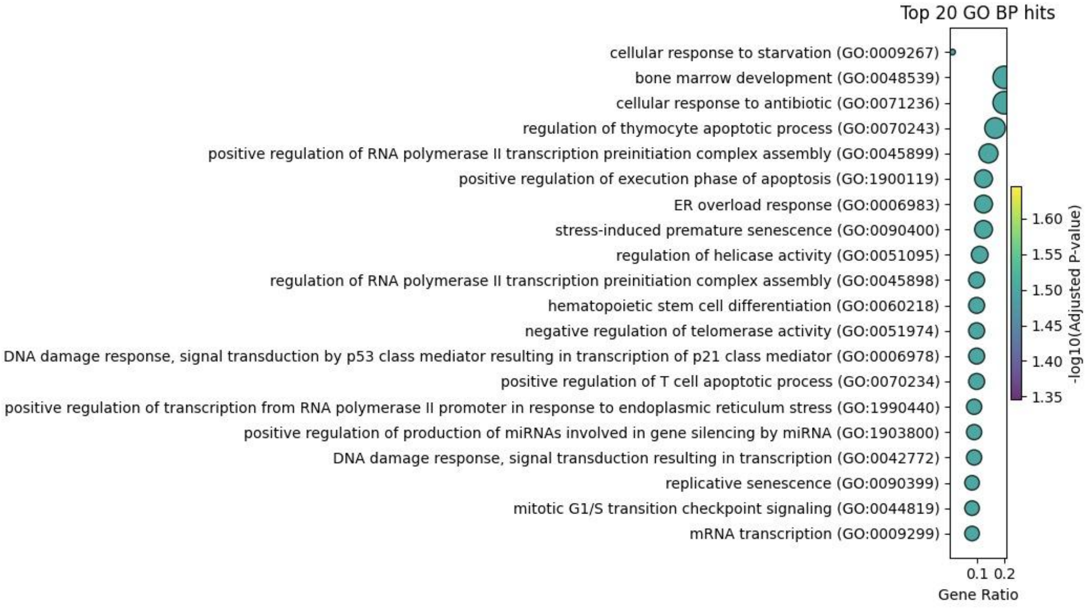
Logistic Regression



SVM

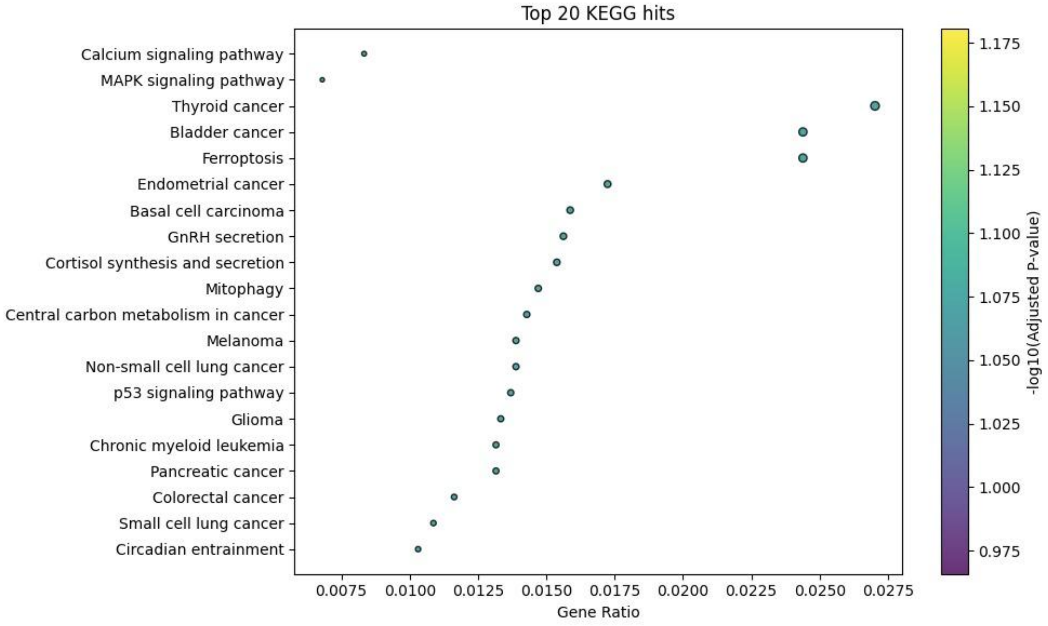
04 Biological Annotation

Rank	GO Term	Adjusted P-value	-log10(P-value)	Key Gene(s)
1	cellular response to starvation	0.031947	2.896538377905295	DAPL1; TP53
2	bone marrow development	0.031947	2.7572101905213233	TP53
3	cellular response to antibiotic	0.031947	2.7572101905213233	TP53
4	regulation of thymocyte apoptotic process	0.031947	2.678194516142461	TP53
5	positive regulation of RNA Pol II preinitiation complex assembly	0.031947	2.6111885865264766	TP53



GO enrichment analysis

Rank	KEGG Pathway	Adjusted P-value	-log10(P-value)	Key Gene(s)
1	Calcium signaling pathway	0.084511	2.5385014732169813	CACNA1I; NTSR1
2	MAPK signaling pathway	0.084511	2.3658251282374	CACNA1I; TP53
3	Thyroid cancer	0.084511	1.8900841369762067	TP53
4	Bladder cancer	0.084511	1.8457586698327018	TP53
5	Ferroptosis	0.084511	1.8457586698327018	TP53



KEGG pathway analysis, several terms were related to **cancer** (e.g., thyroid, bladder) and **signaling pathways** like MAPK and calcium signaling.

TP53 and **CACNA1I** were repeatedly enriched, suggesting they may be key mediators in cancer-related signaling. However, since only 1–2 genes were enriched per pathway, the **adjusted p-values were not significant**, highlighting **limited statistical power**.

KEGG enrichment analysis

TP53 annotation

- TP53 is a key tumor suppressor gene involved in cell cycle control, DNA repair, and apoptosis.

- Frequently mutated in breast cancer, especially in TNBC and HER2 subtypes, TP53 may influence neoadjuvant therapy response.

- In our study, TP53 is enriched in stress-related GO terms (e.g., response to starvation/antibiotics) and KEGG pathways (MAPK, calcium signaling, ferroptosis), suggesting a role in treatment sensitivity.

Summary

Conclusions

- **Logistic Regression & SVM** outperform Random Forest & Gradient Boosting in both evaluation methods
- **Nested Cross-Validation** yields slightly better and more stable metrics than Train/Test Split
- **Top SHAP features**: TP53, ENSG RNA features, CIN.Prop, HER2.status, Grade.pre.NAT, T.stage, DNAH8 mutation
- **Biological Insights:** TP53 appeared frequently in the enriched GO terms, indicating that it is the most functionally central gene in the analysis. The enriched biological processes are mainly related to **cell death**, **drug response**, and **development,** which may be associated with **treatment response** or **tumor aggressiveness**

Challenges & Adjustments

- Small sample size (n = 168), missing validation and testing set
    - Limits statistical power
    - Feature size vs. sample size: required manual feature selection to avoid overfitting
    - Train/test split issues: further splitting the dataset reduces training size and may compromise model stability -> nested cross validation
    - Biological interpretation challenges: Few selected features → pathway enrichment not significant → considered functional annotation instead
- Class imbalance: Addressed using oversampling(SMOTE)
- Data Leakage: missing validation and testing set
        - Acknowledged that feature selection was done before dataset split
        - If time permits, we will split data prior to feature selection to ensure proper validation

Further Work

- **Technical Improvements:**
  - Explore other omics combinations (e.g., Clinical + DNA)
  - Automate annotation for future datasets
- **Biological Validation:**
  - Validate key genes in external cohorts (e.g., TCGA, METABRIC)
  - Assess link to drug targets and resistance
- **Broader Application:**
  - Extend to other cancers with neoadjuvant therapy
  - Collaborate with clinicians for translational use

Thanks for listening!